Exploratory Data Analysis (EDA) Summary Report

1. Introduction

The objective of this Exploratory Data Analysis (EDA) is to analyze customer financial and demographic information to identify patterns and relationships associated with account delinquency. The analysis focuses on variables such as credit utilization, credit score, loan balance, income, employment status, missed payments, and account tenure. Using Excel PivotTables and charts, the dataset is examined to identify potential high-risk indicators, understand delinquency trends, detect anomalies, and assess data quality issues. The findings will help prioritize relevant variables for further investigation and support the development of a reliable delinquency prediction model.

2. Dataset Overview

Key Dataset Attributes:

- **Number of records:** 500 customer records
- **Number of columns:** 21 variables
- **Target variable:** Delinquent_Account (0 = non-delinquent, 1 = delinquent)
- **Duplicate records:** No duplicate rows were identified in the main dataset during the initial review.
- **Missing values:** The original Loan_Balance column contained 29 missing values. The processed new loan balance column contains no missing values following median imputation.

Key Variables and Descriptions:

Variable	Description	Data Type
Customer_ID	Unique customer identifier	Identifier
Age	Customer's age	Numerical
income new	Processed customer income	Numerical
credit score new	Processed credit score	Numerical
Credit Utilization	Proportion of available credit being used	Numerical
customize credit utilization	Categorized credit-utilization level	Categorical
Missed Payments	Number of missed payments	Numerical

Delinquent Account	Indicates whether an account is delinquent	Binary (0/1)
Loan Balance	Original outstanding loan balance	Numerical
new loan balance	Processed loan balance after missing-value treatment	Numerical
Debt_to_Income_Ratio	Ratio of debt obligations to income	Numerical
Employment_Status	Customer's employment category	Categorical
Account_Tenure	Duration of the customer's account	Numerical
Credit_Card_Type	Type of credit card held	Categorical
Location	Customer's recorded location	Categorical
Month_1–Month_6	Monthly payment status, such as On-time, Late, or Missed	Categorical

Initial Observations and Data Inconsistencies:

The initial review identified missing values in Loan_Balance, which were handled using median imputation in the processed column. The dataset also contains inconsistent employment-status labels, including EMP, Employed, and employed, which should be standardized to ensure accurate grouping and comparison.

The main dataset contains no duplicate rows. Numerical variables describe customer demographics and financial characteristics, while categorical variables capture employment, card type, location, credit-utilization categories, and monthly payment behavior. The binary target variable, Delinquent Account, is used to investigate relationships between these characteristics and delinquency outcomes.

The processed columns income new, credit score new, and new loan balance represent transformed variables used during data preparation. The original loan-balance column should be retained for reference and validation.

3. Missing Data Analysis

The dataset was examined to identify missing values and determine appropriate treatment methods to maintain data quality for delinquency analysis.

Key Missing Data Findings

- **Variables with missing values:** The original Loan_Balance column contained 29 missing values out of 500 records (5.8%). The processed columns income new, credit score new, and new loan balance contain no missing values.

- **Missing Data Treatment:** Median imputation was used to handle missing loan balances. A new column, new loan balance, was created, and the missing values were replaced with the median loan balance of 45,776.
- **Justification:** Median imputation was selected because it is less sensitive to extreme values than the mean and helps retain customer records for further analysis.
- **Validation:** The processed new loan balance column contains no missing values, allowing the dataset to be used for PivotTables, charts, and further exploration of delinquency risk indicators.

Treatment of Other Missing Columns:

During data cleaning, missing values were also addressed in the income and credit-score variables by creating the processed columns income new and credit score new. The processed columns contain no missing values.

The original Loan_Balance column contained 29 missing values, which were handled through median imputation in the new column new loan balance. The other variables, including age, credit utilization, missed payments, debt-to-income ratio, employment status, account tenure, credit card type, location, and monthly payment statuses, did not contain missing values in the final dataset.

The processed columns were checked to ensure that no missing values remained before proceeding with PivotTables, charts, and delinquency analysis.

4. Key Findings and Risk Indicators

Correlations Observed Between Key Variables:

The analysis of 500 customer records identified the following patterns and relationships with the Delinquent_Account variable:

- **Credit Utilization:** Customers in the high-utilization category had a delinquency rate of 18.1%, compared with 12.0% in the low-utilization category. This suggests a potential association between higher credit utilization and delinquency, although the very-high-utilization group had a lower rate of 15.2%.
- **Employment Status:** Unemployed customers had a delinquency rate of 19.4%, while retired customers had a rate of 11.5%. Employment status may be a useful risk indicator, subject to further validation and standardization of categories.

- **Credit Card Type:** Business cardholders recorded the highest delinquency rate at 21.3%, followed by student cardholders at 17.9%. Platinum cardholders had the lowest rate at 11.8%.
- **Missed Payments:** Customers with five missed payments had a delinquency rate of 22.1%. However, the overall Pearson correlation between missed payments and delinquency was approximately -0.03, indicating almost no linear relationship.
- **Age:** Customers aged approximately 47–59 had the highest delinquency rate among the four age groups, at 22.4%, suggesting a potential non-linear relationship between age and delinquency.
- **Credit Score:** The highest credit-score quartile had a delinquency rate of 17.6%, compared with 15.2% in the lowest quartile. This unexpected pattern does not follow the commonly expected relationship between higher credit scores and lower delinquency.
- **Account Tenure:** The Pearson correlation between account tenure and delinquency was approximately -0.04, indicating a very weak negative linear relationship. However, delinquency rates varied considerably across individual tenure values.

Unexpected Anomalies Requiring Further Investigation:

1. **Weak relationship with missed payments:** Despite its expected importance in credit-risk assessment, the number of missed payments had almost no linear correlation with the delinquency target.
2. **Unexpected credit-score pattern:** Customers with higher credit scores did not consistently show lower delinquency rates, suggesting a need to validate the target definition and examine other customer characteristics.
3. **Non-linear credit-utilization trend:** Delinquency increased from low to high utilization but decreased in the very-high-utilization category. This could reflect differences in group size or customer characteristics.
4. **Inconsistent monthly payment patterns:** In some months, customers recorded as paying on time had higher delinquency rates than customers recorded as paying late or missing payments. For example, in Month 1, the delinquency rate was 19.8% for on-time customers and 16.5% for customers with missed payments. These patterns warrant investigation into how monthly payment statuses relate to the target variable.
5. **Inconsistent employment categories:** Values such as EMP, Employed, and employed may represent the same employment category and should be standardized to prevent misleading comparisons.

Implications for Delinquency Prediction:

Credit utilization, employment status, credit card type, age, and payment history are potential candidate predictors for further investigation. However, the weak correlations and unexpected patterns suggest that these variables should not be treated as confirmed risk predictors without further testing. The analytics team should validate the target variable, investigate data-quality issues, assess relationships between multiple variables, and evaluate predictive performance before selecting the final features for a delinquency prediction model.

5. AI & GenAI Usage

Generative AI was used as a supporting tool to understand data patterns, explore missing-value treatment options, and interpret potential delinquency risk indicators. The data cleaning, PivotTables, and charts were performed in Excel, with AI assistance used to review and summarize the findings.

Sample prompts used:

- "Summarize key patterns and anomalies in the dataset."
- "Suggest a suitable method for handling missing loan balance values."
- "Identify potential relationships between credit utilization, missed payments, and delinquency."

The AI-generated suggestions were compared with the Excel analysis to identify relevant insights and areas requiring further investigation.

6. Conclusion & Next Steps

The EDA identified potential delinquency risk indicators, including credit utilization, employment status, credit card type, age, and missed payments. However, some relationships were inconsistent with expectations, highlighting the need for further investigation. Missing loan-balance values were handled using median imputation, and PivotTables and charts were used to explore customer patterns.

Recommended Next Steps:

- Standardize inconsistent categorical values and validate data quality.
- Investigate unexpected relationships between customer characteristics and delinquency.
- Perform further statistical analysis to evaluate potential risk indicators.
- Use the validated findings to select relevant features and develop a delinquency prediction model.